ODSUM: Benchmarking Open Domain Multi-Document Summarization

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Abstract

Multi-document summarization (MDS) is a classic task in NLP focusing on generating summaries of multiple input documents. In many real-world scenarios, however, the input documents for MDS need to be first identified and retrieved from a large corpus of documents. Studying MDS in this extended "open-domain" setting with incorporating retrieval has been challenging due to the limited number of high quality benchmarks. This paper introduces two new benchmarks for *open-domain multi-document summarization*, built upon existing MDS datasets. These benchmarks are designed with specific use-cases in mind, featuring queries that reflect real-world scenarios and a diverse collection of documents, thereby closely simulating practical applications. We experiment with several retrieve-then-summarize pipelines, investigating the performance of a wide array of strong retrievers and LMs. Our findings suggest that MDS in the open-domain remains challenging and models often suffer substantial performance loss when retrieval errors happen. Our human evaluation analysis demonstrates that existing automated metrics fail to faithfully reflect the true performance of these models. Moreover, we explore various summarization strategies aimed at mitigating performance losses.

1 Introduction

Multi-document Summarization (MDS) is an established task in NLP aiming to summarize an input consisting of multiple documents (Fabbri et al., 2019b; Lu et al., 2020). In the MDS task, source documents are explicitly assumed and provided. However, in certain realistic scenarios such as aggregating information from news, literature or business, the input resources need to be first identified and retrieved from a large collection of documents, which represent a user information need. In this setting, the summarization performance is highly

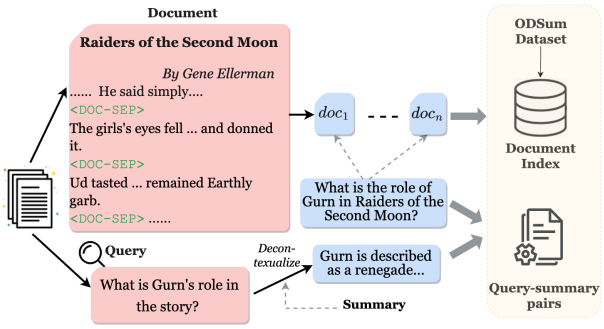


Figure 1: Overview of the creation process of ODSum. The ODSUM benchmark has three distinct features: (1) it requires models to retrieve and aggregate information from a *large document corpus*; (2) it necessitates the combination of data from *multiple documents* to adequately address the queries, simulating complex information retrieval scenarios; (3) it is grounded in *realistic queries* that reflect genuine user information needs, adding practical relevance to the task.

dependent on the set of initial documents identified and retrieved. This setting has been also referred to as MDS in "open-domain" (Ji et al., 2013; Giorgi et al., 2023) (or ODMDS in short). In ODMDS, relevant information to an information need is jointly identified and synthesized from a large corpus of documents, where such information may be disjoint, overlapping, or complementary between multiple documents. The process aims to produce a concise and coherent summary.

ODMDS has close connections with MDS, query-focused MDS, and open-domain QA. While MDS research often involves hundreds of relevant input documents (Wallace et al., 2020; DeYoung et al., 2021b), ODMDS examines a more challenging setting with a large-scale document collection necessitating retrieval. Compared with Query-focused MDS (Xu and Lapata, 2020; Pasunuru et al., 2021) where the input documents are available and already highly relevant to the query, ODMDS requires the initial identification of such documents. While ODMDS shares some parallels

Task	Dataset	Domain	# Examples	Query Source	Avg. Sum. Len.
MDS	DUC 2003-04	News articles	500	-	110
	Multi-News (Fabbri et al., 2019a)	News articles	56,216	-	264
	Multi-Xscience (Lu et al., 2020)	Scientific literature	40,528	-	117
	WCEP	News articles	92,560	-	33
	MS2	Medical studies	415,333	-	58
QMDS	DUC 2005-07	News articles	1,593	Human written	250
	AQuaMuSe (Kulkarni et al., 2020)	Wikipedia	5,519	Rewritten from NQ dataset	106
	QMDSCNN (Pasunuru et al., 2021)	News articles	371,971	Title of news article	250
	QDSIR (Pasunuru et al., 2021)	General web	102,595	Bing search logs	250
	QMSum (Zhong et al., 2021)	Meeting transcripts	1808	Human written	70
	SQuALITY (Wang et al., 2022)	Sci-fi stories	625	Human written	237
ODQA	ASQA (Stelmakh et al., 2022)	Wikipedia	6,316	NQ dataset & Wikipedia	65
	QuALITY (Pang et al., 2022)	Mixed-Genre	381	Human written	11
ODMDS	**ODSUM**	Meetings, Sci-fi stories	1,422	Rewrittn from MDS dataset	244

Table 1: Comparison between ODSum and other existing datasets.

with open-domain QA (Yang et al., 2015; Lewis et al., 2020b; Stelmakh et al., 2022), it focuses on summarizing information from multiple sources given a user's information need, instead of providing short answers.

One challenge in studying open-domain MDS is the lack of realistic benchmarks due to the difficulty of data collection. We aim to bridge this gap, particularly focusing on the limitations observed in recent work on ODMDS (§2). Two primary challenges (Tang et al., 2021; Giorgi et al., 2023) include the use of unrealistic, pseudo-queries (using the reference summary of documents or document titles as the retrieval query) and the sub-optimal construction of the document corpus. To address these limitations, we propose a novel method for data collection and introduce novel benchmarks, ODSUM, tailored to test the limits of ODMDS (§3). Bootstrapped from existing long-context, query-focused MDS datasets, our benchmarks aim to meet the main requirements of a realistic ODMDS task: (1) it necessitates retrieving and aggregating information from a large corpus; (2) the information to address the query must come from multiple documents; and (3) the queries are realistic, reflecting a user's real-world information need (§3).

An efficient ODMDS system should adeptly retrieve relevant information from extensive document collections, accounting for varying quality and relevance. We experiment with a wide range of retrieval and summarization methods, including sparse and dense retrievers, task-specific MDS models, and state-of-the-art LLMs. To evaluate our system, we use standard evaluation metrics and conduct human evaluations.

The effectiveness of our tested approaches in retrieval quality varies significantly. This highlights the need for a more effective solution to the challenges of document quality and relevance variability in ODMDS tasks. Our contributions are summarized below:

- We propose a robust method (§3) to construct ODMDS datasets from qMDS datasets, and present a new benchmark for studying ODMDS.
- We conduct extensive experiments on ODMDS system components, specifically retrieval (§4.1) and summarization (§4.2), providing insights into their capabilities and limitations.
- We observe significant discrepancies among evaluation metrics, with some metrics better capturing the correlation between summarization and retrieval performance, notably G-EVAL (§5.2).
- We investigate various LLM strategies for ODMDS, offering insights into potential improvements (§6).

2 Related Work

Multi-Document Summarization (MDS) MDS amplifies the utility of summarization by synthesizing information from *multiple* various sources (Yasunaga et al., 2017; Liu et al., 2018; Fabbri et al., 2019a; Liu and Lapata, 2019; Li et al., 2020; Lu et al., 2020; DeYoung et al., 2021a). MDS primarily focuses on summarizing information from a set of *provided* documents, without being guided by a specific query. In contrast, ODSum aligns more closely with real-world information-seeking scenarios, where models are required to generate tailored summaries tailored to users' specific information needs. This involves retrieving query-relevant doc-

uments from large corpora and then synthesizing the information into a coherent summary.

Query-Focused MDS (QMDS) QMDS is a specialized form of MDS (Kulkarni et al., 2020; Pasunuru et al., 2021), aiming to generate a short summary from a set of *given documents* that are relevant to a user query. Large-scale high-quality datasets for QMDS are scarce. Benchmarks like AQuaMuSe (Kulkarni et al., 2020) and QMDSIR (Pasunuru et al., 2021) automatically retrieve documents with given queries through mining large corpora or web search engines. However, these datasets only include retrieved documents and lack *gold* related documents. QMDSCNN (Pasunuru et al., 2021) adapts existing datasets into QMDS by using a title as a pseudo-query and four retrieved documents per query as a source. This approach does not guarantee the relevance of the retrieved documents, and the pseudo-query may not capture genuine information needs. Although the document set provided in QMDS datasets may include documents irrelevant to the query, the set is predefined and usually small in size. In contrast, ODSum presents more significant challenges in information retrieval, necessitating models to first retrieve query-relevant documents from a large corpus. This setting closely aligns with real-world information-seeking scenarios, where users typically do not have a predetermined set of documents.

Open-Domain MDS (ODMDS) As previously discussed, both MDS and QMDS assume a set of topic-related documents is provided as input. However, in practice, this document set may not be readily available and needs to be retrieved based on an information need, a scenario we refer to as ODMDS. A closely related task is factoid-based open-domain QA (Stelmakh et al., 2022; Pang et al., 2022), which focuses on generating explanatory answers to factoid questions. However, answers to these questions are shorter than typical expected summaries in MDS and that answers do *not* necessarily come from multiple sources. Recent ODMDS work (Giorgi et al., 2023) uses ground-truth summaries as queries to retrieve relevant documents from a corpus, and generate a broad summary based on the retrieved documents. However, in real-world scenarios, users typically search document corpora for information-seeking purposes. Their questions can be *ambiguous*, posing more challenges in information retrieval compared to using a long and detailed answer as the query. A closer examination of its dataset reveals a discrepancy in handling sparse databases where relevant information is minimal. For example, in one setting the task is to retrieve 10 documents, but on average, only 2.7 of these were relevant. This discrepancy underscores a challenge in handling a sparse database where the majority of retrieved documents do not contain pertinent information. Considering these limitations, there is a clear need for new benchmarks that can address the complexities of ODMDS.

LLMs for Retrieval and Summarization LLMs-based indexes have been found to outperform strong *retrieval* baselines such as BM25 (Robertson et al., 1995) and self-supervised dense retrieval methods (Brown et al., 2020). They are particularly effective in tasks that lack sufficient supervised data (Schick and Schütze, 2020; Winata et al., 2021; Bonifacio et al., 2022). In *summarization* tasks, recent studies (Goyal et al., 2022; Liu et al., 2023c; Zhang et al., 2023; Yang et al., 2023; Zhao et al., 2023) demonstrate that LLMs produce human-preferred and factually accurate summaries when appropriately prompted with a task description. Additionally, LLMs have demonstrated competence in automated summarization evaluation (Liu et al., 2023c; Shen et al., 2023; Mao et al., 2023). We therefore use LLMs (i.e., GPT-* and Llama-2) as strong baselines in our study.

3 ODSUM Benchmark

We initiate our data collection using existing high-quality benchmarks for long-context single-document summarization. Given a source dataset, we segment a long document into shorter chunks, and then filter and rewrite the queries from the dataset to make them decontextualized (stand-alone). Unlike previous work (Giorgi et al., 2023), which used reference summaries as pseudo-queries, our dataset construction ensures that the queries represent realistic information needs. Importantly, our approach preserves the integrity of the gold summaries while making them more applicable in broader open domain scenarios. We detail the dataset construction pipeline as follows:

3.1 Source Data Collection

In this work, we use SQuALITY (Wang et al., 2022) and QMSum (Zhong et al., 2021) as source datasets. *SQuALITY* (Wang et al., 2022) fo-

cuses on question-based summarization of long stories. Expert contractors wrote the summaries in SQuALITY, and in order to ensure quality through cross-annotator reviews, the summary for each question was written by four different annotators. The SQuALITY dataset features 127 stories with 635 questions, leading to a total of 2540 summaries. *QMSum* (Zhong et al., 2021) is designed for query-based summarization of multi-domain long meeting transcripts. The summaries are derived from transcribed meetings, focusing on the extraction and summarization of relevant segments in response to specific queries. The dataset comprises 1,808 query-summary pairs that span across 232 meetings from multiple domains. These pairs were meticulously created, ensuring a wide variety of queries and corresponding summaries. Using these two data sources, we create two new ODMDS benchmarks: ODSUM-STORY and ODSUM-MEET. Table 2 outlines the data statistics of each benchmark. We next detail the benchmark construction process.

3.2 ODSUM-STORY Construction

Stories inherently narrate a sequence of events, and SQuALITY uniquely features narratives with recurring characters, such as character *Retief* in a series. Given that each chapter can stand alone as an independent narrative, we can consider each as a distinct light document. We segment the stories into chapters using natural web delimiters like "`<hr class="***"/>`", in the HTML-scraped forms of the stories. These separate chapters are then considered as individual documents in an open-domain MDS setup. The questions in SQuALITY often pertain to multiple chapters of a story, inherently positioning our dataset as a query-based setup which is convenient for building open-domain MDS datasets.

Decontextualizing Queries When extracted from their original context, the original queries from SQuALITY were naturally vague and required additional context from the document. Therefore they require decontextualization (Choi et al., 2021). We used `GPT-4` (model version: `gpt-4-0613`) to rephrase and decontextualize the queries, enhancing their specificity and relevance. A previous approach (Laskar et al., 2023) generates queries given a document and a summary, but here we rewrite the given query to preserve the exact relevance of the original question to the gold summary as designed in the original datasets, and to make the question standalone so they can be used in real retrieval use case.

Our approach involved two scenarios: one using the original query and title, the other using the query, title, and summary. We listed results in Table 8 in Appendix §A. From these results, we selected the queries that performed the best, modifying them based on both the title and the answer, to create our new query set. For example, we transformed a generic query like "*What is the plot of the story?*" into a more context-specific query such as "*What is the story of CULTURAL EXCHANGE and how does it relate to the harvest of Bacchus vines in Lovenbroy and the arrival of exchange students from Bogan?*". This specificity enriched the queries' practicality and relevance to user intent.

Dataset	**ODSUM-STORY**	**-MEET**
# Documents	1,190	232
Avg. Document Length	808.54	7176.21
# Queries	635	436
Avg. Query Length	10.79	32.89
# Ref Summary per Query	4	1
Avg. Summary Length	273.80	185.17
Training Set Size	250	262
Test Set Size	260	131
Development Set Size	125	43

Table 2: Data Statistics of the constructed ODSUM-STORY and ODSUM-MEET dataset.

Annotation Quality	**%S $\geq$ 4**
Summary Coherence and Fluency	97.5
Open-domain Retrieval Necessity	90.0
Multi-Document Information Aggregation	88.8
Query Relevance and Coverage	83.8
Factual Consistency and Accuracy	85.0
Inter-Document Relationship Capture	82.5

Table 3: Human evaluation over 80 samples (40 from each of ODSum-Meeting and ODSum-Story subsets) of ODSum. Two expert annotators rate the samples on a scale of 1 to 5. The detailed evaluation guidelines are presented in Appendix §F.

3.3 ODSUM-MEET Construction

For QMSum, we leveraged the dataset's thematic continuity and focused discussion characteristic. Each source document in QMSum is a transcript of a meeting, and the meetings typically revolve around persistent themes as participants strive to make progress in discussions. For instance, in the Product setting that includes 137 meetings about

designing a new remote control, the topic of *buttons* recurs in 131 of them, *rubber buttons* in 24, and *colored buttons* in 3. When seeking information on *colored buttons*, an ideal scenario would involve a query specifically about *colored buttons*, with a summary consolidating discussions from the three relevant meetings.

Cluster and Merge Query-Summary Pairs We cluster the original queries in the dataset based on their similarity, subsequently merging similar queries into a single, more comprehensive one. This process extends to summaries, merging them into a single summary that provides a condensed overview of several meetings. The result is a query-summary pair naturally aligned with multiple meetings, providing a rich resource for ODMDS tasks. Details of the process and particular examples can be found in Appendix §B.

3.4 Human Validation of Data Quality

To ensure the high quality and utility of the ODSum dataset, we conducted rigorous human evaluations. We randomly selected 40 examples from each of the ODSUM-MEET and ODSUM-STORY subsets and asked two expert annotators to assess the quality of dataset examples on a scale of 1 to 5 using six criteria. The full annotation criteria and guidelines are presented in Appendix §F.

As shown in Table 3, the vast majority of examples received high scores across all criteria. Over 80% of the examples were rated 4 or higher for each criterion, with Summary Coherence and Fluency, and Open-domain Retrieval Necessity scoring at 97.5% and 90%, respectively. These results corroborate the suitability of ODSum for benchmarking open-domain multi-document summarization systems.

4 Experiment Setup

Retrieve-then-summarize (Liu et al., 2018; Giorgi et al., 2023) is a process where relevant documents are *retrieved* from a large corpora using query and *summarized* into a concise text. We implement and evaluate this pipeline for the ODSUM tasks.

4.1 Retrieval Models

Following Giorgi et al. (2023), we extend the retrieval models to include both *sparse* and *dense* retrievers, as well as *LLM-embedding-based* retrieval, into our framework.

Sparse Retrievers Sparse retrievers calculate the relevance of a document to a query based on overlaps between terms, which are then weighted by their respective frequencies. In this paper we used BM25 (Robertson et al., 1995) as a representative and widely used approach in this family.

Dense Retrievers Dense retrievers embed both documents and queries into a shared embedding space. This provides a more flexible and context-aware approach in measuring the relatedness of text strings. We adopted two different dense retrievers: Contriever (Izacard et al., 2021) and the LLM-based GPT-embeddings by OpenAI.[1]

4.2 Summarization Models

We conduct experiments with a series of summarization models: (See Appendix §E.1 for details). **BART** (Lewis et al., 2020a), **PRIMERA** (Xiao et al., 2022), **GPT Series** (Brown et al., 2020; OpenAI, 2023), and **Llama-2** (Touvron et al., 2023) We list the prompts for LLMs in Appendix §D.

4.3 Retrieval Evaluation

For automated evaluation, we report nDCG (Järvelin and Kekäläinen, 2000), and MAP (Voorhees, 2004). A detailed description can be found in §E.2.1. In addition to these automated metrics, we also performed human evaluation to assess the qualitative aspects of our retrieval system. For the Story domain, we considered **Coverage Percentage** (*Cov%*) to evaluate the proportion of retrieved content originating from the target story, **Relevance Percentage** (*Rel.%*) to measure the pertinence of this content, and **First Valid Position** (*1st Pos.*) to identify the rank of the initial relevant document retrieved. For the Meeting domain, we call a query **answered** if the retrieved content completely addressed the query's information need, and we counted the **Answered Percentage** (*Ans.%*), along with *Rel.%* and *1st Pos.*. Incorporating human evaluation offers a more thorough assessment of our retrieval system, especially in complex narrative and conversational contexts. We provide the guideline for annotators in Appendix §C.

4.4 Summarization Evaluation

We evaluate the quality of summaries generated by different models using the following three metrics: (see Appendix §E.2.2 for details). **ROUGE** (Lin,

[1] https://platform.openai.com/docs/guides/embeddings

	ODSum-story					ODSum-meeting				
Method	NDCG	MAP	Cov%	Rel.%	1st Pos.	NDCG	MAP	Ans%	Rel.%	1st Pos.
BM25	32.49	18.92	54.0	44.7	2.07	**28.14**	**22.71**	41.1	67.1	1.48
Contriever	41.51	24.51	60.1	47.3	1.82	17.40	13.37	47.2	84.0	1.50
GPT-embedding	**48.43**	**30.13**	**69.0**	**52.3**	**1.27**	25.26	19.74	**86.7**	**90.0**	**1.30**

Table 4: Retrieval performance. Here Contriever is a dense retriever and BM25 is a sparse retriever.

2004), **BERTScore** (Zhang et al., 2020) and **G-EVAL** (Liu et al., 2023b), which is a framework that employs LLMs with a chain-of-thoughts (CoT) prompting approach and a form-filling paradigm to assess the quality of NLG outputs. G-EVAL has demonstrated a stronger correlation with human judgments than ROUGE and BERTScore on the SummEval (Fabbri et al., 2021) benchmark (Liu et al., 2023b).

5 Experimental Result and Analysis

5.1 Analysis of Retriever Performance

As shown in Table 4, in ODSUM-STORY, the LLM-based retriever outperforms others, demonstrating superior results in terms of NDCG, MAP and Coverage (Cov%), which reflect its ability to understand narrative context and identify relevant content effectively. Metrics such as Relevance (Rel.%) and First Valid Position (1st Pos.), underscore its edge in that retrieved segments are not just from the same source but also relevant.

In ODSUM-MEET, GPT-embedding also leads in Relevance and First Valid Position. Additionally, our findings draw attention to the Ans% metric in the ODSUM-MEET setting, where the retriever's task is to fully capture the information need of the query. The GPT-embedding method excels in this aspect, outperforming the others with the highest Ans% score. This indicates its proficiency in capturing the complete scope of information in meeting scenarios.

These insights also suggest that while traditional retrieval metrics provide a quantitative measure of retrieval performance, they may not fully capture the effectiveness of retrievers in summarization tasks, as also discussed in §5.2. The dense retrievers' superior summarization performance, despite mixed retrieval metric results, points to a discrepancy between what is measured by retrieval metrics and what contributes to successful summarization.

In our direct evaluation of query-relatedness, the GPT-embedding method consistently retrieves documents more relevant to the query than BM25, particularly in meeting notes where document interrelatedness challenges traditional retrieval methods.

5.2 Summarization Experiments and Analysis

We list the results of summarization in Table 5 and Table 6. The rows are ranked from generally the best performance to the worst.

Discerning Retrieval Errors through Evaluation Metrics We observe that G-EVAL is more reliable for ODSum evaluation, capturing the variances in different retrieval methods. Models show little variation in ROUGE and BERTScore, with GPT-4 performing worse than GPT-3.5 despite being widely regarded as superior (Goyal et al., 2022; Zhang et al., 2022). Comparing the same model using different retrieval methods reveals only slight fluctuations in these scores, and sometimes documents with errors surpass the oracle setting. These findings raise doubts about the reliability of ROUGE and BERTScore in capturing the nuances of summarization quality. In contrast, G-EVAL shows considerable variability across models and retrieval settings, with GPT-4 outperforming other LLMs and pretrained models. G-EVAL also reveals a significant performance gap between the oracle setting and other retrieval methods, demonstrating its ability to provide a more nuanced evaluation landscape for the ODMDS task, especially when considering retrieval errors. Therefore, we analyze the impact of retrieval errors using G-EVAL.

Impact of Retrieval Errors A crucial observation from our experiments is that summarization quality is directly subject to retrieval errors. In both settings (Table 5), the oracle setting, representing an ideal retrieval scenario, outperforms all other retrieval settings (BM-25, Contriever, and *text-ada-002*). The performance ranking of summarization models aligns closely with that of retrieval methods: *text-ada-002* > Contriever > BM-25, suggesting that summarization performance significantly varies based on the quality of retrieved documents, making retrieval a bottleneck for effective summarization.

	BART			PRIMERA			GPT-3.5			GPT-4			LLAMA2 - 70B		
	R-2	BScr	G-EVAL	R-2	BScr	G-EVAL	R-2	BScr	G-EVAL	R-2	BScr	G-EVAL	R-2	BScr	G-EVAL
Oracle	11.57	84.55	20.59	10.63	86.10	20.46	12.28	86.26	39.66	10.24	85.68	46.54	10.75	85.72	35.62
GPT-embedding	11.28	84.63	20.71	10.12	84.56	17.56	10.18	85.19	35.24	6.98	84.46	**38.90**	9.23	85.09	34.64
Contriever	10.84	84.59	19.37	10.53	85.95	19.00	9.34	84.87	31.80	6.76	84.48	36.63	8.63	84.81	32.37
BM-25	10.91	84.46	18.71	10.18	84.23	18.21	9.15	84.91	31.29	5.33	84.21	34.18	8.77	84.78	31.27

Table 5: Summarization performance on ODSUM-STORY. R-2 and BScr stands for ROUGE-2 and BERTScore.

	BART			PRIMERA			GPT-3.5			GPT-4			LLAMA2 - 70B		
	R-2	BScr	G-EVAL	R-2	BScr	G-EVAL	R-2	BS	G-EVAL	R-2	BScr	G-EVAL	R-2	BScr	G-EVAL
Oracle	11.68	86.35	25.65	10.04	85.86	23.85	11.89	85.30	33.68	8.73	84.50	36.39	8.36	85.05	32.26
GPT-embedding	10.86	86.21	23.17	10.46	85.31	23.59	9.88	85.31	31.11	7.48	84.53	**35.32**	8.03	84.82	29.96
Contriever	11.10	86.22	23.67	10.62	84.69	22.10	8.15	84.69	28.18	8.76	84.72	32.60	8.17	84.90	28.28
BM-25	10.83	86.30	24.05	9.63	85.91	22.48	6.96	83.71	24.53	6.01	84.10	25.57	7.32	84.48	27.25

Table 6: Summarization performance on ODSUM-MEET. R-2 and BScr stands for ROUGE-2 and BERTScore.

LLM's Sensitivity to Retrieval Errors GPT-4 experiences a significant decline in performance when retrieval errors are introduced across both story and meeting settings. In ODSUM-STORY, the performance score drops from 46.54 (oracle) to 34.18 (BM-25), and in the meeting setting, from 36.39 to 25.57. `GPT-3.5-turbo-16k` and `LLAMA2-70B` display similar performance in the oracle setting, with `LLAMA2-70B` exhibiting higher robustness, maintaining more of its performance despite retrieval errors.

GPT-4's sensitivity to retrieval errors emphasizes the need to account for real-world deployment context. Despite its superior performance under optimal conditions, its effectiveness can significantly diminish otherwise. This observation further highlights the utility of the G-EVAL, which captures performance nuances overlooked by traditional metrics like ROUGE and BERTScore.

The superior performance of LLMs in the oracle setting and their sensitivity to retrieval errors highlight the importance of effective retrieval in the ODMDS task. Our human evaluation results further validate the utility of the dataset for benchmarking ODMDS systems, suggesting that ODSum presents a challenging and realistic testbed for assessing retrieval and summarization components in an open-domain setting.

6 Methods to improve the performance

To enhance the performance of the MDS and ensure the robustness of LLMs in information retrieval, we conducted ablation studies focusing on the best and worst-performing retrieval settings. We used ODSUM-MEET, contrasting the optimal oracle setting against the least effective sparse setting and the MAX setting to assess their respective impacts on LLM performance. In both scenarios, document lengths exceeded the gpt-3.5-turbo-16k model's maximum context. This prompts us to explore alternative strategies beyond mere input truncation.

6.1 Methods to Improve Robustness

Several methods are investigated to make the LLM more resilient against retrieval errors.

Truncation Methods Due to the context size limitations of LLMs (Appendix §E.1), we employed two truncation strategies to manage document size: (1) *Truncate all:* All retrieved documents are concatenated together and then truncated. The model is left with only the highly ranked documents, which are typically the most relevant according to the retriever's ranking. (2) *Truncate one:* The input consists of documents concatenated from clusters, ordered as they originally appear. Each individual document is truncated by dividing the total context length by the number of documents. This ensures that each document is proportionally represented, which has been found effective in recent work on MDS (Xiao et al., 2022; Caciularu et al., 2023). The truncated parts from each document are then merged by the LLM to form a new document.

Map-Reduce This architecture allows for summarizing large collections of documents by distributing the work across mappers and reducers.

Method	setting	r1	r2	rL	G-Eval
Truncate_all	Oracle	40.20	11.89	20.62	33.68
	with error	*-8.51*	*-4.93*	*-3.10*	*-9.15*
Truncate_one	Oracle	40.03	11.41	20.72	35.42
	with error	*-6.70*	*-3.81*	*-2.48*	*-10.94*
Map-Reduce	Oracle	35.90	10.72	18.90	**37.70**
	with error	*-3.97*	*-3.99*	*-1.72*	*-12.16*
Refine_H2L	Oracle	38.23	10.60	19.48	36.36
	with error	*-4.87*	*-2.51*	*-2.55*	*-10.23*
Refine_L2H	Oracle	42.23	11.19	21.45	34.48
	with error	*-7.37*	*-4.06*	*-4.84*	*-9.26*

Table 7: Different methods to summarize documents with retrieval errors.

The document is divided into smaller chunks. Each of these chunks is initially summarized using an initial prompt (serving as the map function). Subsequently, these intermediate summaries are combined into a final summary using a second, different prompt (serving as the reduce function). This method offers scalability, but there is a possibility of losing information during the intermediate step.

Refine Method This method iteratively updates the generated summary by sequentially processing each retrieved document. In each round, the current document along with the latest interim summary is input into an LLM to generate a revised summary. This approach is also scalable, but the order in which the documents are processed may impact its performance. Two types of document orders can be considered: ranking highest to lowest or vice versa, based on relevancy. While truncation methods offer speed, they may compromise depth. The Map-Reduce approach scales well but may lose details in the intermediate steps. The Refine method, although thorough, can be expensive and is influenced by document order.

6.2 Discussion

The results presented in Table 7 provide insights into different strategies for improving the robustness of LLM against retrieval errors, particularly in the context of the ODSUM-MEET dataset. In the oracle setting, the *Truncate_all* method performs slightly better than *Truncate_one*, and *Map-reduce* achieves the highest score among all the settings, suggesting that summarizing or truncating all documents to maintain information diversity may be more beneficial than focusing only on the initial, supposedly more relevant documents. Although *Map-reduce* performs the best in terms of G-EVAL, it has the lowest ROUGE score. This can be attributed to the method's two-level abstraction of word selections and the introduction of more novel words. This result supports our observation in §6 that the ROUGE score does not adequately reflect summary quality.

Both variants of the Refine Method show interesting results. *Refine_H2L* (High to Low relevance) performs well in G-EVAL, while *Refine_L2H* (Low to High relevance) achieves the highest ROUGE score in the oracle setting. This suggests that the order in which documents are processed is critical, as the model is likely to prioritize words from the most recent document.

Methods that include more information in the model generally achieve better scores when retrieval is ideal, implying that providing more information benefits the model's summarization capabilities. However, when retrieval errors are introduced, these advanced methods experience a significant performance drop and suffer more than simpler truncation methods. These findings suggest the need to explore methods that maximize relevant information inclusion while mitigating the negative impact of retrieval errors. Truncation methods may have a "natural resilience" to these errors due to their constrained operation, limiting exposure to irrelevant information. Advanced methods should incorporate similar safeguarding strategies to improve robustness.

7 Conclusion

This paper introduces a novel dataset, ODSUM, for the ODMDS task. The experimental approach adopted in this study covers both the retrieval and summarization stages within the "retrieve-then-summarize" pipeline. This approach allows us to identify areas that require further investigation in the ODMDS task. During our analysis, we observed significant variations among the evaluation metrics used. Specifically, we found that certain metrics, such as G-EVAL, effectively highlight the correlation between summarization and retrieval performance. Furthermore, we studied the effectiveness and robustness of various summarization strategies, including truncation, map-reduce, and refine, providing insights into their individual advantages and limitations. Our human evaluation demonstrates the quality and utility of the ODSUM dataset, making it suitable for benchmarking open-domain multi-document summarization systems.

Limitation

Concurrent with our work, recent advancements have developed LLMs capable of handling inputs up to 256K in length and beyond (Tworkowski et al., 2023) or even virtually unlimited length (Bertsch et al., 2023). However, our study focuses on the "retrieve-then-summarize" pipeline without exploring end-to-end methodologies that process long-context datasets directly to generate responses. We investigate this approach because it allows for more controlled and interpretable processing of information (Liu et al., 2023a), ensuring that critical data is accurately captured and summarized. This aspect is particularly important in domains where precision and reliability are paramount.

We also reflect on the limitations inherent in our ODSUM design and potential biases introduced by using LLM-generated queries. Despite efforts to ensure quality, the complexity and scale of our dataset, combined with the use of automated query construction, may introduce inaccuracies. The inclusion of a large number of documents aimed to simulate realistic open-domain settings inevitably introduces a level of noise and that poses a challenge in evaluating the ODMDS systems.

Our evaluation setup that relies on OpenAI GPT-4 raises concerns regarding potential biases. Despite the controlled nature of these biases within our comparative analysis, more comprehensive manual meta-evaluation in future work is imperative to ensure the alignment with human judgment.

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Method	Top-k Strategy	P@K	R@K
Original	min(3)	0.48	0.25
	mean(8)	0.36	0.34
	max(10)	0.32	0.33
With title	min	0.66	0.35
	mean	0.49	0.46
	max	0.42	0.45
With title and answer	min	**0.74**	0.40
	mean	0.54	**0.52**
	max	0.47	0.50

Table 8: Performances using BM25 with different modifications of queries.

Appendix

A q2OD-MDS

Here we present a mathematical definition for q2OD-MDS:

qMDS: $\{(q_i, D_i, s_i)\}_{i=1}^n$

where:

q_i is the i-th query,

D_i is the set of documents related to q_i,

s_i is the summary related to q_i.

Transformation:

Step 1: Extract query-summary pairs:

$\{(q_i, s_i)\}_{i=1}^n$

Step 2: Combine all documents: $D = \bigcup_{i=1}^{n} D_i$

ODMDS: $(D, \{(q_i, s_i)\}_{i=1}^n)$

B Transforming QMSum

We formalize the process of transforming QMSum into a new dataset mathematically as follows:

- **Meetings Transcripts:** Denote the set of all meetings transcripts as $T = \{T_1, T_2, \ldots, T_n\}$, where T_i represents the transcript of the i-th meeting.
- **Queries & Summaries:** Each meeting transcript T_i has an associated set of queries $Q_i = \{q_{i1}, q_{i2}, \ldots, q_{im}\}$, where q_{ij} represents the j-th query in the i-th meeting. Each query q_{ij} has a corresponding summary s_{ij}. Therefore, we have a set of summaries $S_i = \{s_{i1}, s_{i2}, \ldots, s_{im}\}$ for each meeting transcript T_i.
- **Query Clustering:** Denote the function that measures the similarity between two queries as $\text{sim}(q_{ij}, q_{kl})$, where q_{ij} and q_{kl} are queries from meetings i and k, respectively. Here, we used *text-embedding-ada-002* for the purpose of implementing $\text{sim}(q_{ij}, q_{kl})$. We then define a threshold θ such that if $\text{sim}(q_{ij}, q_{kl}) > \theta$, we consider q_{ij} and q_{kl} to be in the same cluster. The result of this step is a new set of query clusters $C = \{C_1, C_2, \ldots, C_p\}$, where each cluster $C_k = \{q_{ij} | \text{sim}(q_{ij}, q_{kl}) > \theta \text{ for all } q_{kl} \text{ in } C_k\}$.
- **Cluster Modification:** We ensured that each cluster has a minimum and maximum size, which can be useful for ensuring that each cluster is meaningful (not too small) and manageable (not too large). Here we choose max to be 6 and min to be 2. For any cluster C_k with more than max queries, we performed a secondary clustering operation within C_k using a lower similarity threshold. Similarly, for any cluster C_k with less than min queries, we merge it with the most similar other cluster. After this step, we have a modified set of clusters $C' = \{C'_1, C'_2, \ldots, C'_q\}$.
- **Merging:** For each modified cluster C'_k, a new merged query Q'_k and a corresponding summary S'_k are created by combining the queries and summaries in the cluster, respectively. Here we also asked LLM to merge them together for us. After this step, we have a set of merged query-summary pairs $\{(Q'_1, S'_1), (Q'_2, S'_2), \ldots, (Q'_q, S'_q)\}$, where each pair corresponds to a modified cluster.

The final result of this process is a new set of query-summary pairs $\{(Q_k, S_k)\}$, providing a condensed overview of multiple meetings.

We show one particular example here.

Query list:

- `What did the group discuss about the function of rolling through the user's favorite channels?`
- `What did the group discuss about the object of the remote control?`
- `What did the group discuss about the functional Remote control?`
- `What function did the group think should be on the remote control?`
- `What did the group discuss about the functions of the remote control?`

- How did they discuss the kinetic function of the remote control?

Result:

- What did the group discuss about the function, object, and kinetic function of the remote control?

C Guideline for Annotators

Foreword

We need to do the assessment of the retrieval process inherent in the document index.

In oracle setting , each query is paired with its source documents,. In contrast, a "retrieved-setting" is a scenario where documents are matched with retrieved counterparts, primarily on the axis of similarity.

In ODSUM specifically, we can select the oracle setting against the "LLM-embedding setting". Using a randomized selection to have 50 samples that lie between these two settings. Use the mean setting in LLM-embedding setting.

Dataset Quality Evaluation

STORY

Here the goal is to evaluate the retrieval process of the document index.

Step 1: Read the Line and Split the Document

Begin by reading from the database. Each will contain a query and a document separated by <doc-sep>. Split the document into different story parts using the <doc-sep> delimiter.

Step 2: Count the Number of Story Parts

Count the number of story parts obtained from the split in Step 2. This will give you the result for Requirement 1.

Step 3: Identify Story Parts in the Same Story

Go through each story part and determine how many of them belong to the same story as indicated by the entire document. This might involve checking for continuity in the narrative or recurring characters or themes. Tally the number of story parts that fit this criterion. This will give you the result for Requirement 2.

Note that in oracle setting it should always be all of the story parts.

Step 4: Identify Relevant Story Parts

Using the query from the line, determine how many of the story parts (that are in the same story) are relevant to the query. Tally the number of relevant story parts. This will give you the result for Requirement 3.

Step 5: Determine the Position of the First Relevant Story Part

Identify the position (in ranking) of the first story part that was found to be relevant in Step 5. For instance, if the third story part was the first one that was relevant, your result for this would be 3. This will give you the result for Requirement 4.

Step 6: Record the Results

For each line of the database, record your findings in the following format:

- Number of story parts (from Requirement 1)
- Number of story parts in the same story (from Requirement 2)
- Number of relevant story parts (from Requirement 3)
- Position of the first relevant story part (from Requirement 4)

You can use a simple table or spreadsheet to record these results.

MEETING

Step 1: Read the Line and Split the Document

Begin by reading from the database. Each entry will consist of a query and a document separated by <doc-sep>. Split the document into different meeting segments using the <doc-sep> delimiter.

Step 2: Count the Number of Meeting Segments

Compute the number of meeting segments obtained from the split in Step 1. This will cater to Requirement 1.

Step 3: Identify Relevant Meeting Segments

Utilizing the query from the line, determine how many of the meeting segments (that are part of the same meeting) are relevant to the query. Count the number of relevant meeting segments. This will cater to Requirement 2.

Step 4: Determine the Position of the First Relevant Meeting Segment

Pinpoint the position (in terms of ranking) of the initial meeting segment that was deemed relevant in Step 3. For instance, if the second meeting segment was the first relevant one, your outcome for this step would be 2. This meets Requirement 3.

Step 5: Determine if Query information not answered

In meeting setting, it is possible that there are many aspects of information is asked. The aim is to discern whether the retrieved meeting segments or

story parts contain details does not fully cover the requested by the query. This meets Requirement 4.

Step 6: Document the Results

For each line of the database, document your discoveries in the following structure:

- Number of meeting segments (from Requirement 1)
- Number of relevant meeting segments (from Requirement 2)
- Position of the first relevant meeting segment (from Requirement 3)
- The retrieved documents can fully reply the query or not (from Requirement 4)

A straightforward table or spreadsheet would be ideal for recording these results.

D Prompt Templates

D.1 Summarization Prompt Templates

- **ODSUM-STORY**:
  You are a helpful assistant that gives long answer to question based on a long story. Write an answer based on the following question and the story. STORY:{story} QUESTION:{query} SUMMARY:
  Here *STORY* represents a selection of the most matching stories retrieved by different retrieval models based on the query from ODSUM-STORY.

- **ODSUM-MEET**:
  You are a helpful assistant that gives long answer to question based on a long meeting. Write an answer based on the following question and the given meeting. Try to answer thoroughly and do not leave out useful information. MEETING:{meeting} QUESTION:{query} SUMMARY:
  In this case, *MEETING* refers to a selection of the most matching meetings retrieved by different retrieval models based on the queries from ODSUM-MEET.

D.2 G-EVAL Prompt Templates

- **G-EVAL consistence:**
  You will be given one prediction summary and one reference summary. Your task is to rate the prediction on one metric. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed. Evaluation Criteria: Consistency (1-5) - the factual alignment between the prediction and the reference. A factually consistent prediction contains only statements that are entailed by the reference. Annotators were also asked to penalize predictions that contained hallucinated facts.
  Evaluation Steps: 1. Read the reference carefully and identify the main facts and details it presents. 2. Read the prediction and compare it to the reference. Check if the prediction contains any factual errors that are not supported by the reference. 3. Assign a score for consistency based on the Evaluation Criteria.

- **G-EVAL relevance:**
  You will be given one prediction summary and one reference summary. Your task is to rate the prediction on one metric. Please make sure you read and understand these instructions carefully. Please keep this document open while reviewing, and refer to it as needed. Evaluation Criteria: Relevance (1-5) - selection of important content from the source. The prediction should include only important information from the reference. Annotators were instructed to penalize predictions which contained redundancies and excess information.
  Evaluation Steps: 1. Read the prediction and the reference carefully. 2. Compare the prediction to the reference and identify the main points of the reference. 3. Assess how well the prediction covers the main points of the reference, and how much irrelevant or redundant information it contains. 4. Assign a relevance score from 1 to 5.

E Models

E.1 Summarization Models

BART (Lewis et al., 2020a) is a sequence-to-sequence Transformer model pre-trained using a mixture of denoising objectives and is considered a strong baseline for summarization tasks. We use a checkpoint[2] of `BART-Large` fine-tuned on the CNN/DailyMail dataset (Hermann et al., 2015). The model has a maximum context length of 1024

[2] https://huggingface.co/facebook/bart-large-cnn

tokens. We fine-tune the model on ODSUM using the AdamW optimizer (Loshchilov and Hutter, 2019) for a maximum of 6 epochs with early stopping. Following previous work (Wang et al., 2022), we format input examples by concatenating a query q to the beginning and the end of the document, separated by an added `[SEP]` special token. For ODSUM-STORY, every (document, query) input corresponds with four training instances, one for each reference. Due to its limited context length, we consider this model as a simple baseline.

PRIMERA (Xiao et al., 2022) is a long-context transformer model designed for MDS tasks by pre-training on task-inspired pre-training objectives and achieves strong results for MDS on both zero/few-shot and full fine-tuning setting. We fine-tune the PRIMERA model on each setting of our dataset. Because the max input token length is 4K, we truncate each retrieved document to fit in the context limitation.

GPT Series (Brown et al., 2020; OpenAI, 2023) are large language models introduced by OpenAI and demonstrate state-of-the-art results on majority of the language-related tasks. We utilize both `gpt-3.5-16k-turbo-0613` and `gpt-4-0613`[3] to perform the summarization task. We find that in order to get high performance, we need careful design of appropriate prompts to correctly instruct the GPT models to carry out the task. Throughout this process, after designing and testing a variety of prompts, we found that GPT models generally prefer concise, shorter prompts and usually perform better in such scenarios. We also observed that when a query is placed at the end of an article, the resulting summary tends to be of higher quality. Our prompts are listed in Appendix §D. We truncate stories/meetings to suitable lengths to fit within the maximum input tokens of GPT-3.5 (16k tokens) and GPT-4 (8k tokens).

Llama-2 (Touvron et al., 2023) is an open-source LLM that has exhibited versatility ranging from logical reasoning to text generation. We use the largest available model, the 70 billion-parameter `Llama-2-70b-Chat` checkpoint, which was optimized for following instructions. At inference time, we load the model weights in 4-bit using NF4 quantization (Dettmers et al., 2023).

Input Format for BART and PRIMERA The input to BART and PRIMERA models consists of the top-k retrieved story segments forODSUM-STORY and the top-k retrieved meeting transcripts for ODSUM-MEET. We use model-corresponding special tokens to denote the separation between each document, ensuring clear demarcation within the input data.

Token Limits of Models

- BART: 1,024 tokens
- PRIMERA (using Longformer-Encoder-Decoder (Beltagy et al., 2020)): 4,096 tokens
- Llama2: 4,096 tokens
- GPT-3.5-turbo-16k: 16,385 tokens
- GPT-4-0613: 8,192 tokens

E.2 Evaluation Models

E.2.1 Automated Retrieval-Evaluation models

nDCG (Järvelin and Kekäläinen, 2000) (Normalized Discounted Cumulative Gain) is a metric quantifying the quality of ranking in retrieval systems, and MAP (Voorhees, 2004) (Mean Average Precision) provides a single-figure measure of quality across recall levels. Each metric offers a different perspective on the system's performance, helping us understand its strengths and weaknesses.

E.2.2 Summarization-Evaluation models

ROUGE (Lin, 2004) a classic metric for summarization evaluation, measuring the word overlap between the candidate and reference summaries. We report ROUGE-2 F1 score.

BERTScore (Zhang et al., 2020) calculates the similarity between the reference and generated summary using contextual word embeddings. We chose F1 score as the metric for BERTScore.

G-EVAL (Liu et al., 2023b) is a framework that employs LLMs with a chain-of-thoughts (CoT) approach and a form-filling paradigm to assess the quality of NLG outputs. Given the original text, questions, and generated answers, the GPT model is instructed to provide scores across various dimensions (e.g., consistency, coherence, relevance, and fluency) based on the given prompts. G-EVAL is able to achieve better performance than ROUGE and BERTScore across different dimensions of summarization evaluation on the SummEval (Fabbri et al., 2021) benchmark.

[3] https://platform.openai.com/docs/models/overview

G-EVAL Implementation Details We used `gpt-3.5-turbo-16k-0613` as the backbone of G-EVAL. GPT-4 is reportedly better in the evaluation, but given the amount of data we're processing here it is too cost-ineffective and we leave it for future work. Here we compare the consistency and relevance between the predicted summary and the reference summary. Consistency measures whether the information in the predicted summary is faithful to the reference summary, while relevance measures the semantic similarity between them. We note that different from the original implementation of G-EVAL, we use G-EVAL for *reference-based* evaluation because the input documents can exceed the maximum processing length of GPT-3.5. Besides, it enables a fair comparison with ROUGE and BERTScore which are reference-based. The prompts we used can be found in appendix §D.

F Human Validation Criteria and Guidelines

Summary Coherence and Fluency
Measures the overall language quality and coherence of the summary.

- **5:** The summary is coherent, well-organized, and easy to read.
- **3:** The summary is mostly understandable but has some issues with coherence and clarity. The flow may be somewhat disjointed and the language a bit difficult in places.
- **1:** The summary is very difficult to understand due to severe issues with coherence and organization.

Open-domain Retrieval Necessity
Assesses the extent to which the query requires retrieving information from a large corpus (*>5*), as opposed to a narrow set (*<3*) of pre-selected documents.

- **5:** The query requires searching a large open-domain corpus to find relevant information.
- **3:** The query could potentially be answered using a focused set of documents, but searching a larger open-domain corpus provides additional relevant information.
- **1:** The query could be fully answered using a small pre-defined set of documents.

Multi-Document Information Aggregation
Measures how well the summary synthesizes and aggregates information from multiple retrieved documents to answer the query. This also reflects the effectiveness of merging summaries by the LLM used in our study.

- **5:** The summary effectively aggregates and synthesizes information from multiple retrieved documents into a coherent answer to the query.
- **3:** The summary uses some information from multiple documents but mainly relies on one or two dominant documents to answer the query.
- **1:** The summary fails to aggregate information from multiple documents at all.

Query Relevance and Coverage
Evaluates how well the summary captures the key information requested in the query.

- **5:** The summary directly and comprehensively addresses all parts of the query, covering the most relevant information.
- **3:** The summary partially addresses the query, covering some relevant information but missing other important aspects.
- **1:** The summary does not address the query or capture any of the key relevant information.

Factual Consistency and Accuracy
Assesses the factual alignment between the summary and the retrieved source documents.

- **5:** The summary is completely factually consistent with the information provided in the retrieved documents.
- **3:** The summary is mostly factually consistent with the source documents, but contains some minor inconsistencies.
- **1:** The summary has severe factual inconsistencies with the source documents, including inaccuracies, contradictions or unsupported (hallucinated) claims.

Inter-Document Relationship Capture
Evaluates how well the summary captures and expresses the relationships between information across different retrieved documents.

- **5:** The summary well reflects the relationships between relevant information from different documents.
- **3:** The summary captures some relationships between information from different documents.
- **1:** The summary does not capture or express any relationships between information from different documents.

F.1 Annotation Guideline

1. Read the query carefully to understand the information needed.
2. Read through the documents to assess their relevance to the query.
3. Read the summary and proceed to evaluate the entire example on the 6 criteria above, assigning a score from 1-5 for each criterion.
4. For each score, provide a brief justification noting key observations and examples from the summary and documents.
5. Add further comments if there is any.

G Dataset Example

We show one example of each setting of the ODSUM dataset in Table 9 and Table 10.

Field	Content
Query	What is the role of Hank Arapoulous in the story Cultural Exchange?
Retrieved_document_1	...What can I do for you? Retief said. You work for this Culture bunch, do you? Funny. I thought they wereall ribbon-counter boys. Never mind. I'm Hank Arapoulous. I'm a farmer.What I wanted to see you about was... He shifted in his chair. Well, out on Lovenbroy we've got a serious problem. The wine crop is justabout ready. We start picking in another two, three months. Now I don'tknow if you're familiar with the Bacchus vines we grow...? No, Retief said. Have a cigar? He pushed a box across the desk. Arapoulous took one. Bacchus vines are an unusual crop, he said,puffing the cigar alight. Only mature every twelve years. In between,the vines don't need a lot of attention, so our time's mostly our own.We like to farm, though. Spend a lot of time developing new forms. Apples the size of a melon...
Retrieved_document_2	Retief lay on his back in deep grass by a stream, eating grapes. A tallfigure appeared on the knoll above him and waved. Retief! Hank Arapoulous bounded down the slope and embraced Retief,slapping him on the back. I heard you were here, and I've got news for you. You won the final day's picking competition. Over two hundredbushels! That's a record! Let's get on over to the garden. Sounds like the celebration's aboutto start. In the flower-crowded park among the stripped vines, Retief and Arapoulous made their way to a laden table under the lanterns. A tallgirl dressed in loose white, and with long golden hair, came up to Arapoulous. Delinda, this is Retief today's winner. And he's also the fellow that got those workers for us...
Retrieved_document_3	The secretary placed the papers on the desk. Arapoulous caught her eyeand grinned. She sniffed and marched from the room. What that gal needs is a slippery time in the grape mash, Arapoulous observed. Retief thumbed through the papers, pausing to read from time to time. He finished and looked at Arapoulous. How many men do you need for the harvest, Hank? Retief inquired. Arapoulous sniffed his wine glass and looked thoughtful. A hundred would help, he said. A thousand would be better. Cheers. What would you say to two thousand? Two thousand? Retief, you're not fooling? I hope not. He picked up the phone, called the Port Authority, asked for the dispatch clerk. Hello, Jim. Say, I have a favor to ask of you. You know that contingent of Bogan students. They're traveling aboard the two CDT transports. I'm interested in the baggage that goes with the students.Has it arrived yet? Okay, I'll wait...
Summary_1	Hank Arapoulous is first described as a bucolic person from Lovenbroy.He is a farmer, tall with bronze skin and gray hair, who comes to MUDDLE's office to discuss the harvest problems in Lovenbroy. They grow Bacchus vines, which only mature once every twelve years. This year is a harvest year, but they don't have enough people to harvest the grapes. Arapoulous explains to Retief that a few years ago, Boge landed a force on Lovenbroy to try to mine their minerals by strip-mining. Lovenbroy fought back for a year but lost a lot of its men. This created financial problems, so Lovenbroy borrowed money from Croanie, mortgaging its crops. The loan is due, and the wine crop will cover the loan amount, but they don't have enough people to harvest the grapes. He is worried that if they don't have a great harvest, Croanie will come in and start mining. Also, if they default on the loan, Croanie will hold half of the grape acreage that they used to secure the loan. Arapoulous has also asked for help from the Labor Office, but they only offered to send them machinery, and machines cannot harvest the grapes. He returns to see Retief the following day to find out if Retief has discovered a way to help. When Mr. Karsh makes a scene about the missing luggage for the exchange students, Retief has Arapoulous take Karsh away and take care of him. When they return, Karsh is stumbling and needs support to stand up. Arapoulous explains that Karsh fell. Retief sends the exchange students to Lovenbroy with Arapoulous to help with the harvest. As the harvest is winding down, Arapoulous tells Retief that Retief has won the award for the picking competition. Arapoulous is also the person who judges the wine contest.
Summary_2	...

Table 9: Example from the ODSUM-STORY Dataset

Field	Content
Query	What were the changes needed according to the group and the Professor's opinion on the use of English in the discussion about IBM computers and data collection participants?
Retrieved_document_1	...People always say very glibly that if you show improvement on a bad system , that doesn't mean anything , cuz it may not be show , because , you know , it doesn't tell you anything about the good system . You know , that if some people If you 're actually are getting at something that has some conceptual substance to it , it will port . professor b: And in fact , most methods that people now use were originally tried with something that was not their absolute best system at some level . If we 're getting three percent error on, English , native speakers , using the Aurora system , and we do some improvements and bring it from three to two , do those same improvements bring, you know, the SRI system from one point three to you know, to point eight ? professor b: You know , that 's that 's pretty solid , on the segmentation stuff . And the Aurora folks here will will definitely get something in on Aurora , phd d: which is not phd f: Actually this this, So, there 's another paper. phd f: And he tested it mostly on digits because it 's sort of a you know , it doesn't take weeks to train it. And got some very impressive results, with, you know , discriminative , Gaussian training. you know, like, error rates go from I don't know , in very noisy environment , like from , ...
Retrieved_document_2	...without the adaptation and compare to these numbers without the adaptation. but I 'm not so much worried about the adaptation , actually , than the , VTL estimation . phd f: If you have only one utterance per speaker you might actually screw up on estimating the the warping , factor . But it 's not the amount of speakers , it 's the num it 's the amount of data per speaker. grad e: So , although I sort of know how to run it , there are a little a f few details here and there that I 'll have to dig out . phd f: And there 's a script and that is actually all in one script . So there 's this one script that parses waveform names and extracts things like the , speaker , ID or something that can stand in as a speaker ID . So , we might have to modify that script to recognize the , speakers , in the in the , TI - digits database. phd f: Or you can fake you can fake names for these waveforms that resemble the names that we use here for the for the meetings . phd f: That would be the , sort of probably the safest way to do grad e: I might have to do that anyway to to do because we may have to do an extract to get the amount of data per speaker about right . grad e: The other thing is , isn't TI - digits isolated digits ? phd f: Right . grad e: Or is that another one ? I looked through a bunch of the digits t corp corpora , and now they 're all blurring . phd f: By the way , I think we can improve these numbers if we care to compr improve them by , not starting with the Switchboard models but by taking the Switchboard models and doing supervised adaptation on a small amount of digit data collected in this setting .
Summary	The script for extracting speaker ID information needs to be modified in order to improve the accuracy of data estimation per speaker. The professor believes that removing English did not significantly impact the model, and it would be valuable to train on one language and test on another. The team expressed concerns about the availability of computational resources, but the professor informed them that they had received two 550 megahertz processors from IBM instead of the expected eight 800 megahertz processors. Currently, there are already 30 subjects lined up for the study, with more expected to be recruited off campus. It was agreed that having subjects select tasks from different categories, such as "big place" or "service", would provide a better range of data. Each dialogue is expected to last no more than 10 minutes. As the data collection is about to begin, minor changes need to be made to the experiment design, script, and permission forms. Subjects can be recruited from within the university or through other social circles.
BART_generated_Summary	The group discussed the possibility of using IBM computers for data collection, with a focus on the performance of the H T K system, which was used by all participants in Aurora, Colorado. They also discussed the use of English as a language in the data collection process, with the professor expressing concerns about the potential loss of English due to lack of usage. However, the professor emphasized the importance of incorporating English into data collection and emphasized the need for the team to learn English properly...
gpt-3.5_generated_Summary	...

Table 10: Example from the ODSUM-MEET Dataset